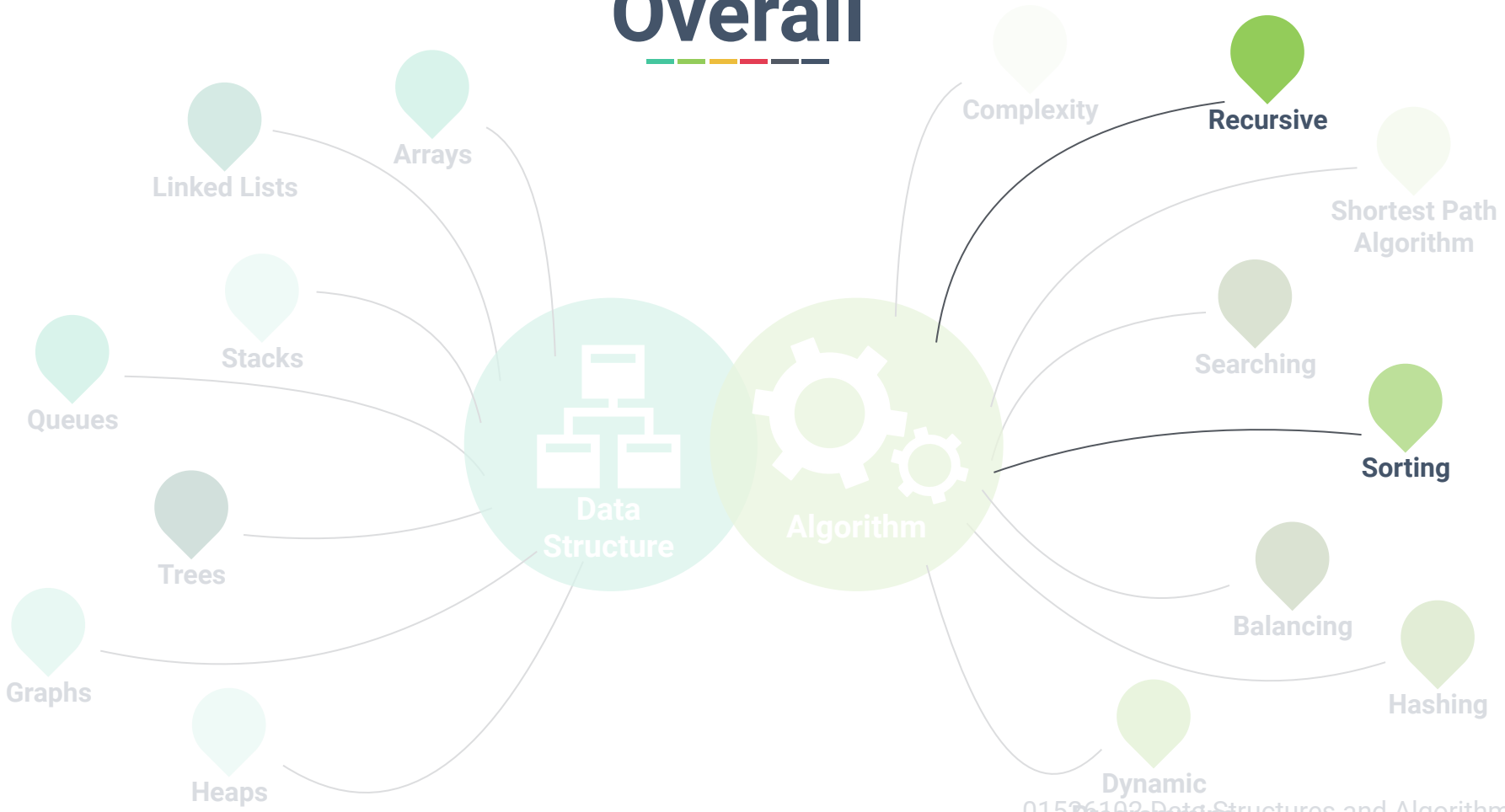
Chapter 10: Recursion & Sorting

Dr. Sirasit Lochanachit

Overall


Linked Lists
Arrays
Stacks
Queues
Trees
Graphs
Heaps
Data Structure
Algorithm
Complexity
Recursive
Shortest Path Algorithm
Searching
Sorting
Balancing
Hashing
Dynamic Programming

Outline

Recursion

- Components of Recursion
- Summation
- Exponentiation
- Factorial

Sorting (Cont.)

- Merge Sort
- Quick Sort

What is Recursion?

- It breaks down a large problem into smaller subproblems that are simpler to solve[1].
- A function that is calling itself one or more times[2].

[1] Bradley N. Miller, David L. Ranum, Problem Solving with Algorithms and Data Structures using Python, 2011
[2] Michael T. Goodrich et al., Data Structures and Algorithms in Python, 2013

What is Recursion?

Ref: https://www.smbc-comics.com/comic/recursion

Recursion

Real-life example:

- Fractal patterns in art and nature
- **Matryoshka** doll
  - Also known as Russian doll or nested doll
  - Set of wooden dolls where smaller dolls are placed inside another.
  - A symbol of motherhood and fertility.

Image Source: https://images.theconversation.com/files/163337/original/image-20170330-4592-1n4ji0f.jpg, https://cdn.pixabay.com/photo/2017/06/13/13/45/matryoshka-2398918_960_720.jpg, https://tf-cmsv2-smithsonianmag-media.s3.amazonaws.com/filer/1d/ff/1dffc703-583f-43d9-a645-96870f5a1d7b/ice-queen.jpg

When to use recursion?

- For problems that contain smaller instances of the same problem.
- To repeat a computer program, **while-loop** and **for-loop** can be used.
- Alternatively, recursion repeats by calling a function itself one or more times.

[2] Michael T. Goodrich et al., Data Structures and Algorithms in Python, 2013

Image Source: https://cdn.pixabay.com/photo/2017/06/13/13/45/matryoshka-2398918_960_720.jpg

Components of Recursion

- Base Case: The smallest subproblem that can be solved easily.

- Recursive Case: A subproblem that reduces the size of the input and move toward the base case

Summation

- Summing list elements recursively.

Summation

- Summing list elements recursively.

Time Complexity: *O*(?)

```
def getSum(data):
  """ Return the sum of the first
n numbers of sequence data."""
  totalSum = 0
  for i in data:
    totalSum = totalSum + i
  return totalSum
```

```
def getSum(data):
  """ Return the sum of the first n
numbers of sequence data."""
  if len(data)==0:
      return 0
  else:
      return data[0]+
getSum(data[1:])
```

Power

Power function takes 2 numbers as input and return the value of *base* to the power of *exp*.

$$\text{power}(base, exp) = \begin{cases} 1 & \text{if } exp = 0 \\ base * \text{power}(base, exp-1) & \text{otherwise} \end{cases}$$

```
def getPower(base,exp):
  """ Return the multiplciation of  base with exp times. """
  if exp==0:
    return 1
  else:
    return base*getPower(base, exp-1)
```

Factorial

Factorial function takes a number as an input and returns the factorial of that number

- n! = Product of positive integers from 1 to n
- If n = 0, then n! = 1 by convention

factorial(1)

factorial(2)

factorial(3)

factorial(4)

factorial(5)

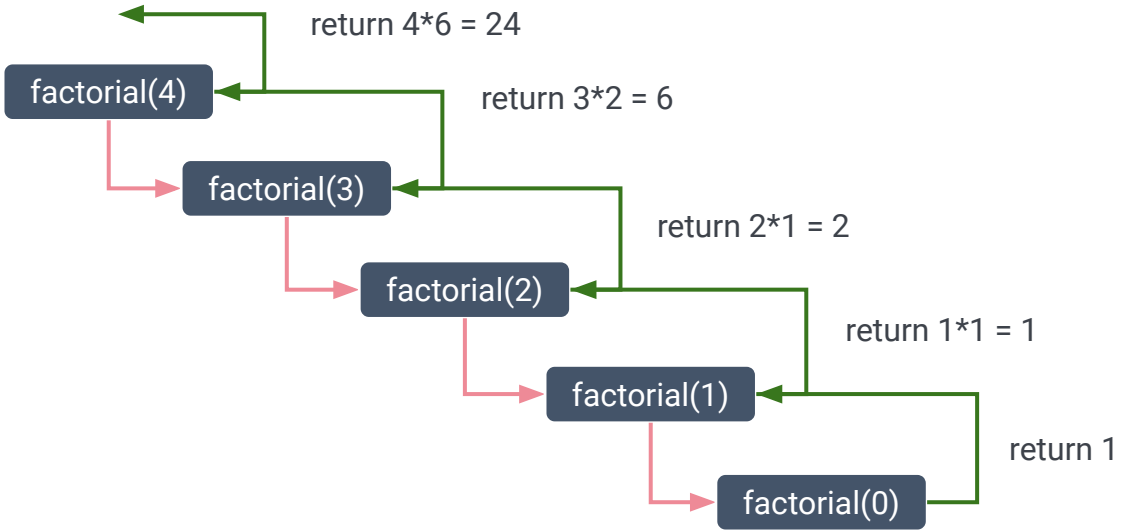
Factorial

Factorial function takes a number as an input and returns the factorial of that number

- n! = Product of positive integers from 1 to n
- If n = 0, then n! = 1 by convention

Recursion

- For recursive, repetition is conducted by repeatedly invoke the function call.



Recap

- Components:
  - Base case and Recursive case
- Examples
  - Summation
  - Power
  - Factorial
  - Binary search
- One rule for recursion: there must be a stop condition.

Image Source: https://cdn.pixabay.com/photo/2017/06/13/13/45/matryoshka-2398918_960_720.jpg

Divide and Conquer Strategy

3 Steps[2]:

1. **Divide**: Divide the input data into two or more disjoint subsets.
2. **Recur**: Recursively solve the subproblems associated with the subsets.
3. **Conquer**: Take the solutions to the subproblems and “merge” them into a solution to the original problem.

[2] Michael T. Goodrich et al., Data Structures and Algorithms in Python, 2013

Merge Sort

Merge sort is a recursive sorting technique based on divide and conquer technique by continually splits a list into equal halves[1].

Base case: If the list is empty or has only one item, it is sorted.

Recursive case: If the list has > 1 item, split the list and recursively merge sort on both halves.

After the two halves are sorted, then a merge is invoked, combining them together into a single sorted new list.

[1] Bradley N. Miller, David L. Ranum, Problem Solving with Algorithms and Data Structures using Python, 2011

Merge Sort Real-life Demo

Merge-sort with Transylvanian-saxon (German) folk dance

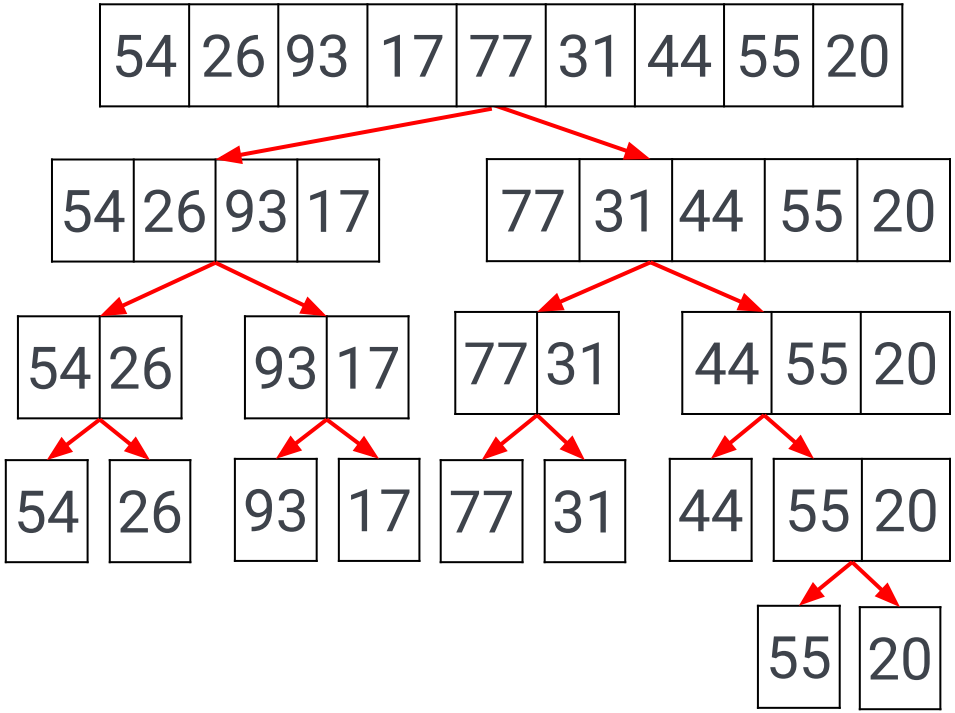
Merge Sort Example

2	45	6	9	15	12	7	8

Merge Sort Example

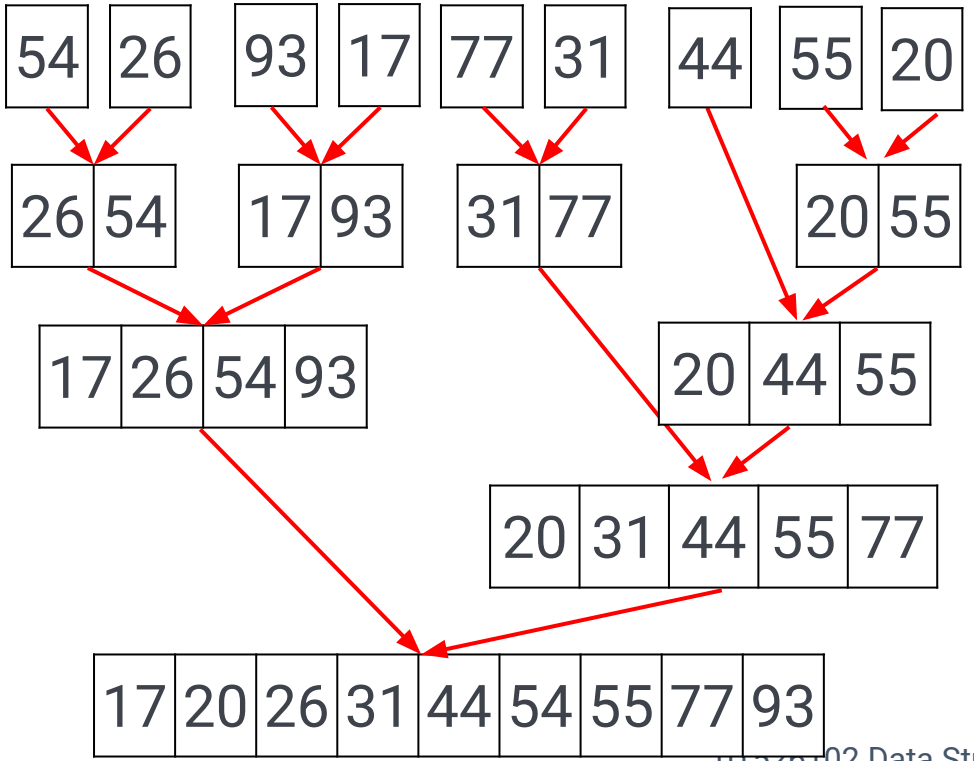
Merge Sort Example

Odd elements



Merge Sort Example

Odd elements



Merge Sort Performance

2 Processes[1]:

1. Splitting the list into halves
2. Merge operation and sorting

Merge sort requires extra memory space to hold two halves as they are continually splitting.

- This can be a critical factor if the list is large.

[1] Bradley N. Miller, David L. Ranum, Problem Solving with Algorithms and Data Structures using Python, 2011

```
Algorithm mergeSort(list):
    if length(list) ≤ 1:
        return list
    else:
        middle = length(list) / 2
        divide list into two unsorted sublists, left and right, using middle
        left = mergeSort(left)
        right = mergeSort(right)
        result = merge(left, right)
        return result
    end if
End mergeSort
```